LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 6 Test (Set B): Control and Coordination

Chapter / Topic: Control and Coordination

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.

SECTION A

6 marks

Choose the correct option.

Q1. Which part of the brain controls voluntary actions? **[1]**

(a) Medulla (b) Spinal cord
(c) Pons (d) Cerebrum (forebrain)

Q2. Which of these is an example of chemotropism? **[1]**

(a) A shoot bending towards light
(b) A pollen tube growing towards the ovule
(c) Roots growing downwards
(d) Leaves of touch-me-not folding

Q3. Growth hormone is secreted by the: **[1]**

(a) thyroid gland (b) adrenal gland
(c) pituitary gland (d) pancreas

Q4. Reflex arcs are formed in the: **[1]**

(a) brain (b) spinal cord
(c) heart (d) muscles

Q5. Which hormone brings about the changes in girls at puberty? **[1]**

(a) Testosterone (b) Insulin
(c) Oestrogen (d) Adrenaline

Q6. **Assertion (A):** We can taste food with our tongue.
Reason (R): Gustatory receptors detect taste, while olfactory receptors detect smell. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. What is the function of receptors in our body? What could happen if they do not work properly? **[2]**

OR

How is a reflex action different from walking?

Q8. How does chemical coordination take place in plants? **[2]**

SECTION C

6 marks

Q9. Name the gland: **[3]**

(i) found on top of each kidney. **[1]**

(ii) in the neck, which needs iodine to make its hormone. **[1]**

(iii) that produces digestive enzymes as well as a hormone. **[1]**

Q10. How are involuntary actions different from reflex actions? Give one example of each. **[3]**

OR

What is the role of the hypothalamus? Explain with the example of growth hormone.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) Describe the structure of a neuron and how information travels through it. **[3]**

(ii) Give two limitations of using electrical impulses for coordination. **[2]**

OR

(a) Describe three types of tropic movements in plants, with an example of each. [3]
(b) Explain why a shoot bends towards light. [2]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. Riya's grandfather has diabetes. His doctor explained that his pancreas does not secrete enough insulin, so his blood sugar level stays high. He has to take insulin injections, and his sugar level is checked regularly to decide the dose. **[4]**

(i) Which gland produces insulin? **[1]**

(ii) What happens to the blood sugar level when insulin is not secreted in proper amounts? **[1]**

(iii) Why do diabetes patients need insulin injections? How does the body normally control the amount of insulin released? **[2]**

~ All the best! ~